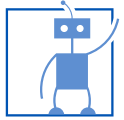


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Reinforcement Learning: Introduction, Fundamentals, Applications

Prof. Dr.-Ing. habil. Alois Knoll

Kejia Chen, M.Sc.

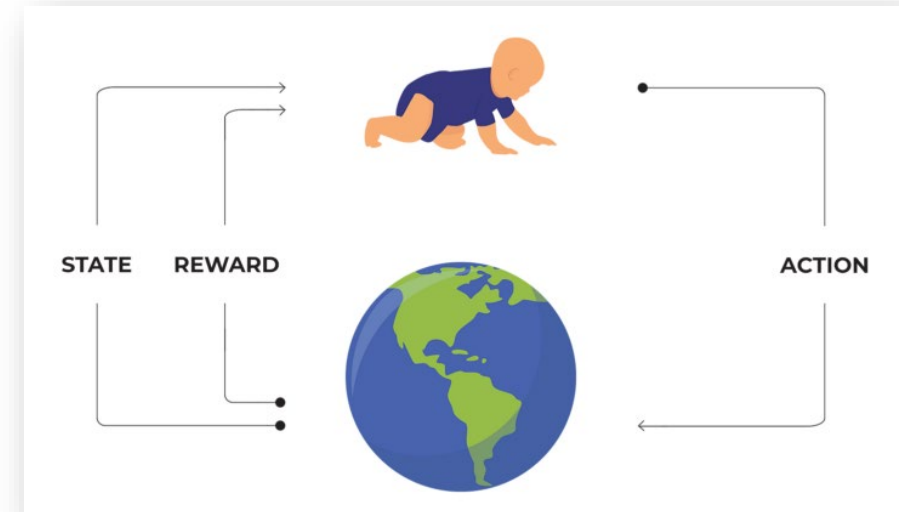
Mohammadhossein Malmir, M.Sc.

Agenda

- Introduction
- Markov processes
- Value Functions
- Generalized Policy Iteration (GPI)
- Dynamic Programming (DP)
- Monte Carlo (MC)
- Temporal Difference (TD)
- Overview & Comparison
- Approximative RL

Interaction Loop

- Humans and animals learn from interaction with our environment without examples.[#]
- The learning is goal-directed.
- Examples of RL:
 - A new-born gazelle that learns running at 20 miles per hour within 30 minutes.[§]
 - Investing across multiple assets (portfolio optimization).*
 - Playing chess, backgammon, video games.*
 - Learning robotic manipulation tasks.*



Alexey Poddiachyi - towardsdatascience

[#]DeepMind x UCL RL Lecture Series - Introduction to Reinforcement Learning, 2021, 1st lecture. **slide 13.**

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.

*David Silver. *Reinforcement Learning: UCL Course*, 2015, 1st lecture. **slide 9.**

RL Multidisciplinary[§]

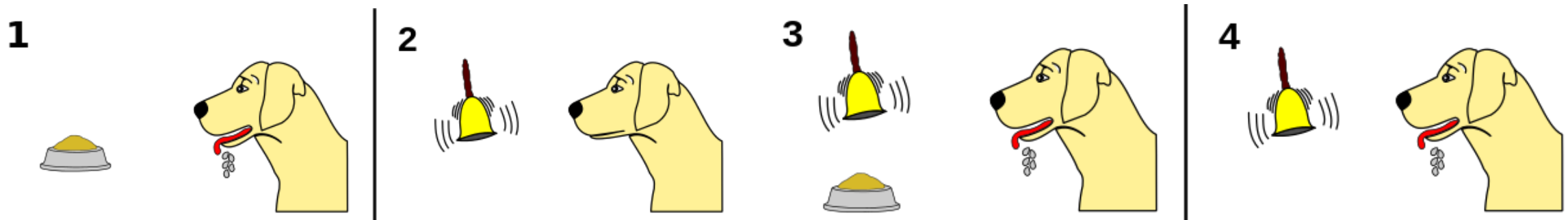
	psy- chology	computer science	neuro- science	math- ematics	engi- neering	economics
reinforcement learning	class./ operant cond.	machine learning	reward system	operations research	optimal control	bounded rationality

[§]adapted from David Silver. *Reinforcement Learning: UCL Course, 2015, 1st lecture. slide 6.*
Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

Learning in Psychology[§]

Classical and operant conditioning are two basic **associative** learning processes that link sensory stimuli to corresponding actions. Both forms of conditioning are very **common** in nature and present in most vertebrate animals.

- **Classic Conditioning:** In classical conditioning, a subject learns the **relationship** between an initially neutral **conditioned stimulus** (CS) and an **unconditioned stimulus** (US) that **reflexively** produces a **conditioned response** (CR). When the subject is repeatedly presented the CS und US together, the CS will finally produce the CR even in the absence of the US.



[§]adapted from Binder, M. D., Hirokawa, N., and Windhorst, U. *Encyclopedia of neuroscience*. Berlin and London: Springer, 2009

Slide is adapted Dr.Ing. Noah from CogSys21 RL lecture held by Prof. Klarmann.

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- **Operant/Instrumental Conditioning:** In operant conditioning, a subject learns the **relationship** between a **stimulus** and its **behavior**. A stimulus is only presented in response to a certain action of the subject and serves as a **reinforcer** that increases or decreases the probability of that action.

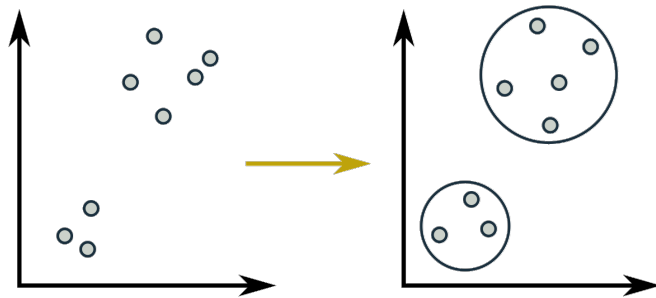
[§]*adapted from Binder, M. D., Hirokawa, N., and Windhorst, U. Encyclopedia of neuroscience. Berlin and London: Springer, 2009*
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RL Multidisciplinary[§]

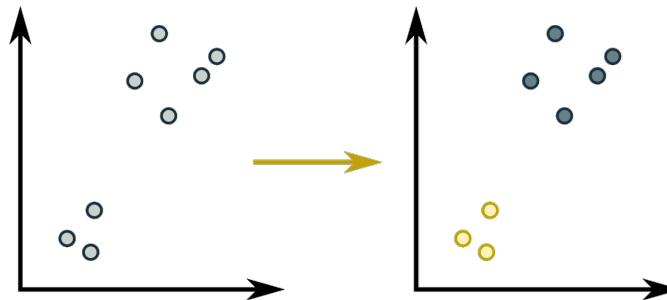
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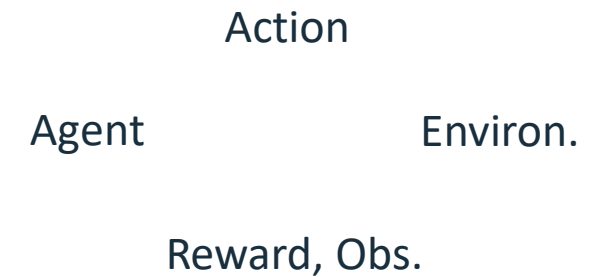
Types of Machine Learning



Unsupervised Learning



Supervised Learning



Reinforcement Learning

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AlphaGo

- Google's Deep Mind created AlphaGo, an agent able to play the game of Go
 - Combines techniques like reinforcement learning, supervised learning, tree search
- AlphaGo was able to win against grand master Lee Sedol in 2016



Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

The Ancient Chinese Game of Go

- GO is an ancient Asian board game
 - it is seen as sacred due to its complexity
 - professionals dedicate their entire life for mastering Go
- Number of states exceeds $1e+80$
 - strategies are based on recognizing abstract patterns
 - players often argument to rather rely on intuition than on strategies



Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

Move 37

- The extraordinary move 37 of AlphaGo made history
- What seemed to be a mistake paved the way of a new style of playing Go



Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

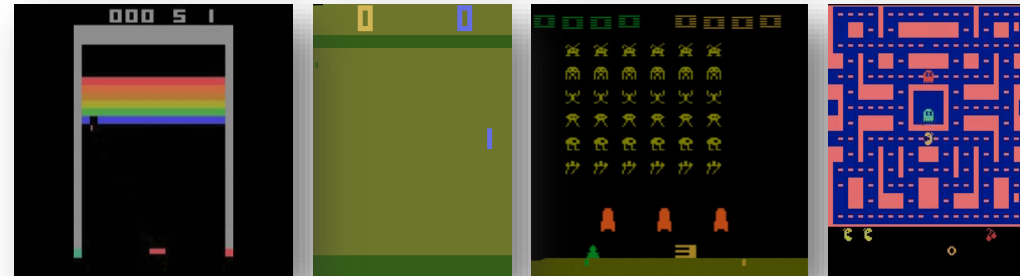
Further Milestones

OpenAI's Five 2018[#]



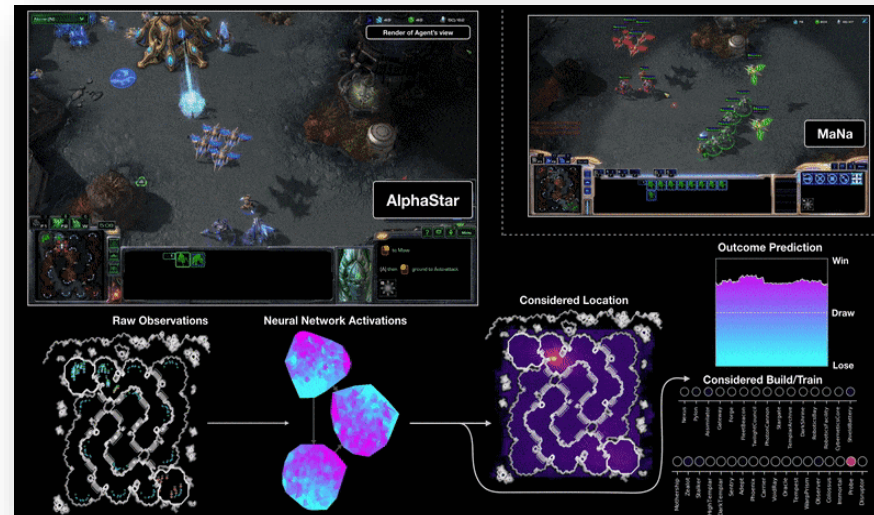
[#]Berner, C., Brockman, G., Chan, B., Dota 2 with Large Scale Deep Reinforcement Learning. URL : <http://arxiv.org/pdf/1912.06680v1>.

Atari™ Games 2014[§]



[§]Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, et al. "Human-level control through deep reinforcement learning". In: Nature 518.7540 (2015), pp. 529–533.

StarCraft II 2019^{*}

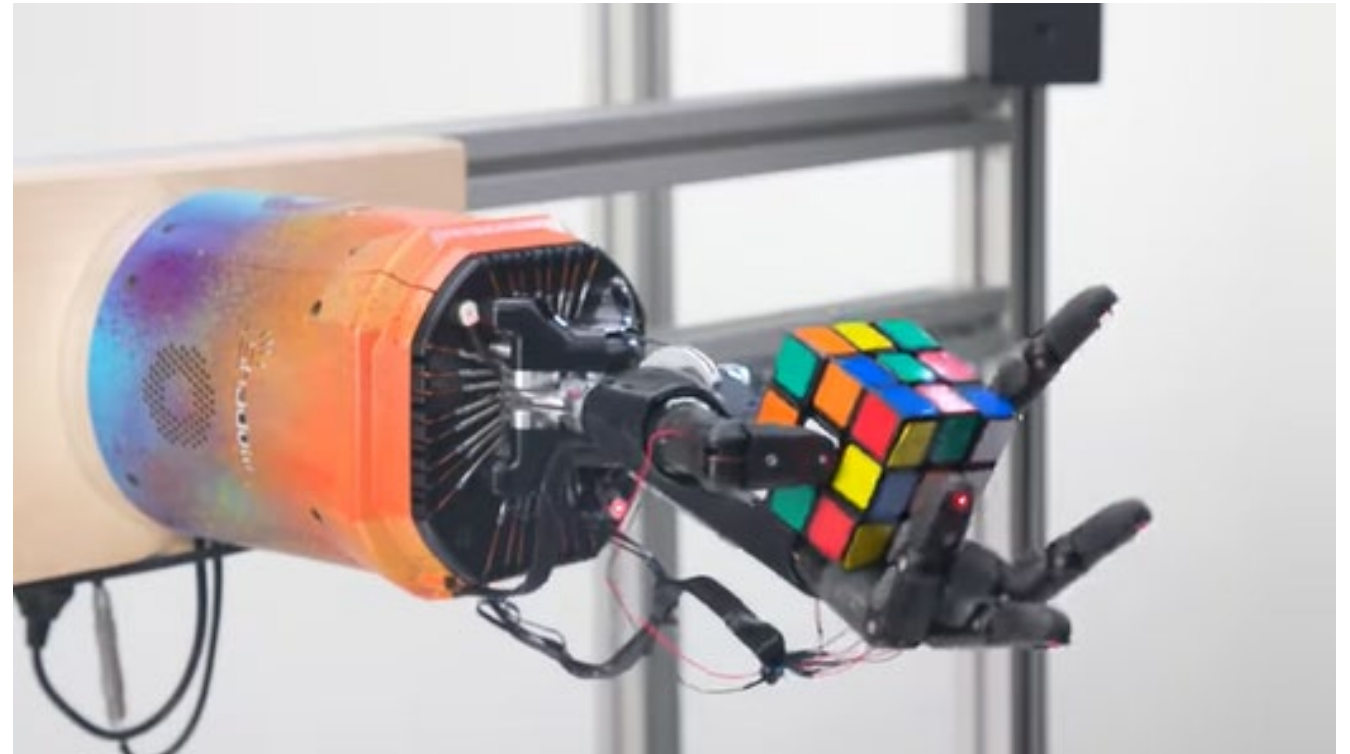


^{*}Vinyals, O., Babuschkin, I., Czarnecki, W. M., et al. "Grandmaster level in StarCraft II using multi-agent reinforcement learning". In: Nature 575.7782 (2019), pp. 350–354.

Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

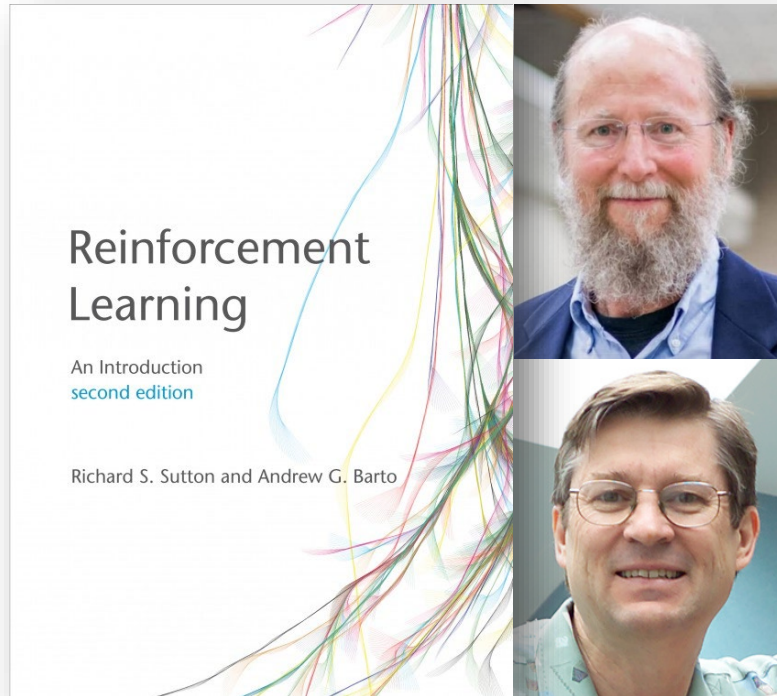
Solving Rubik's Cube with a Robot Hand

- OpenAI solved the Rubik's Cube with a human-like robot hand.
 - reinforcement learning paired with automatic domain randomization



Akkaya, Ilge, et al. "Solving rubik's cube with a robot hand." arXiv preprint arXiv:1910.07113 (2019).

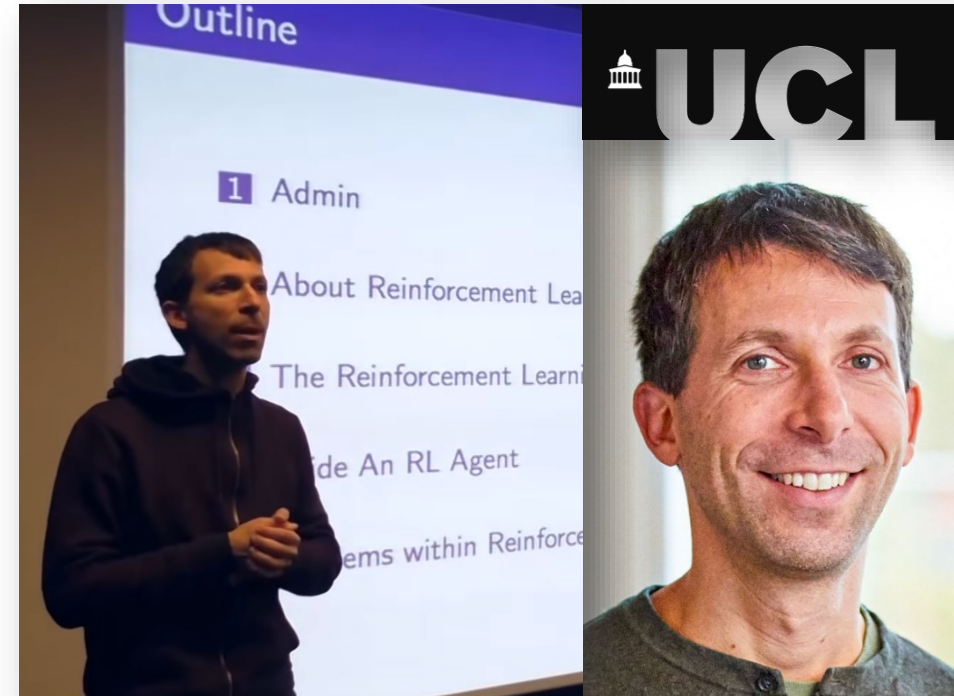
Key References



the full story[§]

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning.* Cambridge, Massachusetts: The MIT Press, 2018.

<http://www.incompleteideas.net/book/RLbook2020.pdf>



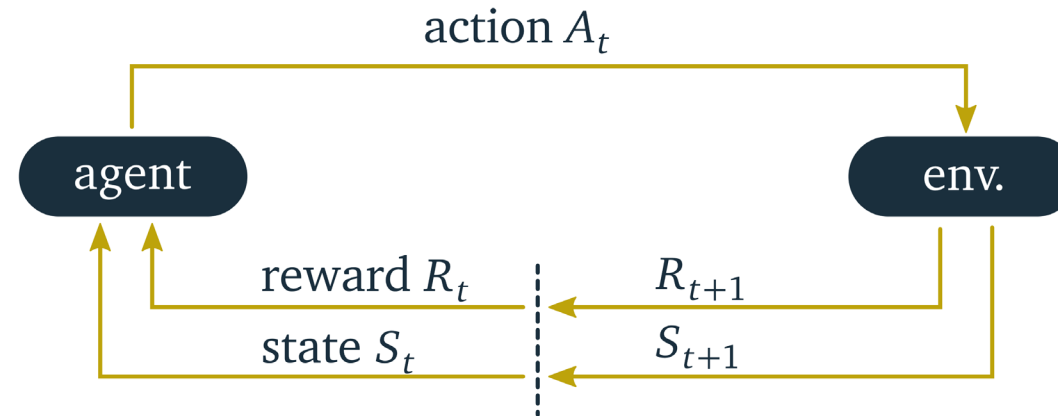
the short cut^{*}

^{*}David Silver. *Reinforcement Learning: UCL Course, 2015.*

<https://www.youtube.com/playlist?list=PLqYmG7hTraZDM-OYHWgPebj2MfCFzFObQ>

Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

Mathematical Formulation



- How should the agent select actions according to the state? We need to evaluate a **policy** describing which action to choose conditioned by the state:

deterministic $a = \pi(s)$

stochastic $\pi(a|s) = \mathbb{P}[A_t = a | S_t = s]$

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.48,58.

Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

π	policy
A	stochastic action function
a	particular action
R	stochastic reward function
S	stochastic state function
s	particular state

Reward Hypothesis

- Reinforcement learning is based on the reward hypothesis:

*That all of what we mean by goals and purposes can be well thought of as the maximization of the expected value of the **cumulative sum** of a received scalar signal (called reward) [§].*

- We introduce **Return** G_t as we usually interested in the cumulative reward

$$G_t := R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

G	return
R	stochastic reward function
S	stochastic state function

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.57.

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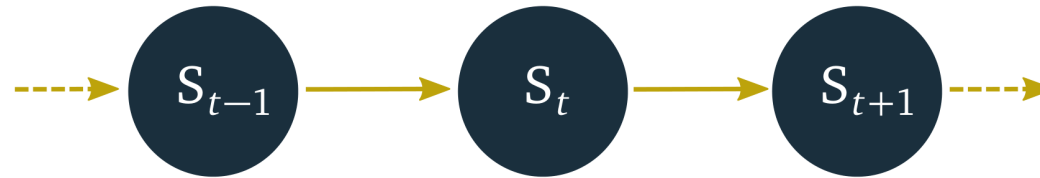
$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=t+1}^T \gamma^{k-t-1} R_k = R_{t+1} + \gamma G_{t+1}$$

the discount rate determines the present value of future rewards.

γ	discount rate
G	return
R	stochastic reward function
S	stochastic state function

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.57.

Markov Processes 1/3[§]



- A sequence of states is described by $\{S_1, S_2, S_3, \dots\}$.
- The sequence is a **Markov Process** $\langle \mathcal{S}, \mathcal{P} \rangle$ if the probability of the next state solely depends on the predecessor state (*Markov property*):

$$\mathbb{P}[S_{t+1}|S_t] = \mathbb{P}[S_{t+1}|S_1, \dots, S_t]$$

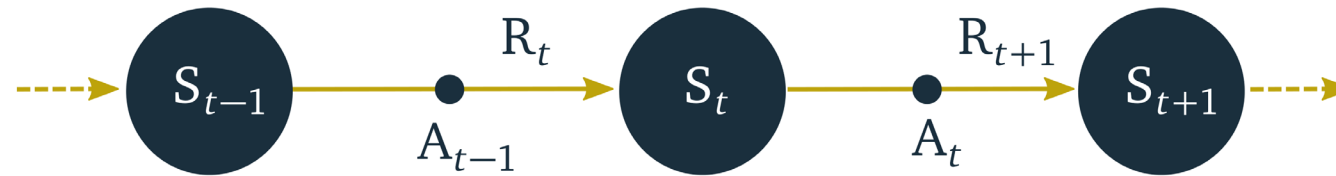
$\mathcal{S}$	finite set of states
$\mathcal{P}$	state transition probability matrix

S	stochastic state function
s	particular state

[§]David Silver. *Reinforcement Learning: UCL Course, 2015*, 1st lecture. **slide 9**.

Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann.

Markov Processes 2/3[§]



- In **Markov Decision Processes** $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, actions steer states (and rewards) in a desired direction.
- The dynamics of MDP:

$$p(s', r | s, a) := \mathbb{P}[S_t = s', R_t = r | S_{t-1} = s, A_{t-1} = a]$$

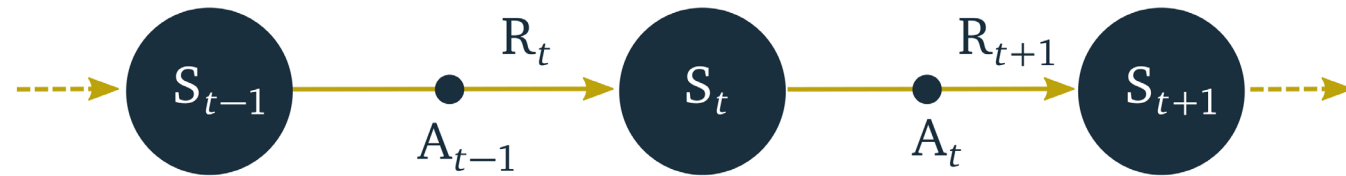
$\mathcal{S}$	finite set of states
$\mathcal{A}$	finite set of actions
$\mathcal{P}$	state transition probability matrix
$\mathcal{R}$	reward function
γ	discount rate

A	stochastic action function	p	transition probability
a	particular action	R	stochastic reward function
S	stochastic state function	r	expected reward
s	particular state		

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.48,49**.

Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann

Markov Processes 3/3[§]



- Based on dynamics, one can compute anything regarding the environment:

$$r(s, a) := \mathbb{E}[R_t | S_{t-1} = s, A_{t-1} = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r | s, a)$$

$$p(s' | s, a) := \mathbb{P}[S_t = s' | S_{t-1} = s, A_{t-1} = a] = \sum_{r \in \mathcal{R}} p(s', r | s, a)$$

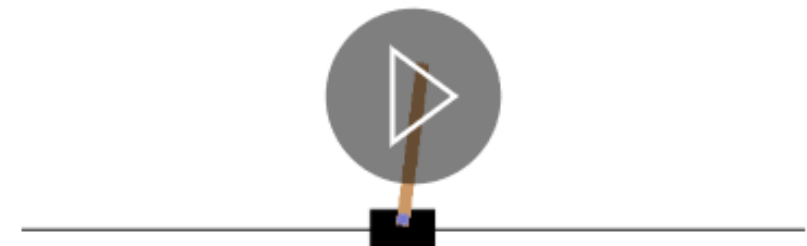
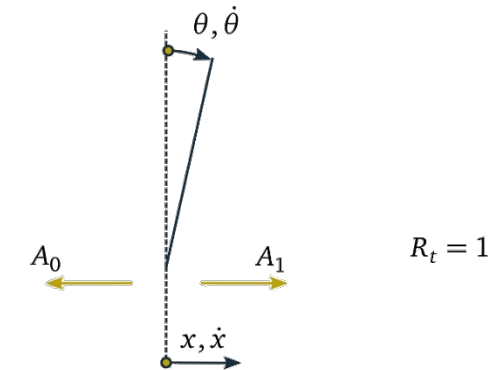
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MDP Example: Cart Pole Environment[§]

- **actions:** apply a force of +1 or -1 to accelerate the cart
- **reward:** +1 for every time step that the pole remains upright
- **return:** number of steps until termination
- **states:** position and speed of the cart; angle and angular velocity of the pole
- **terminal:** pole more than 48° from vertical, cart moves more than 2.4 units from the center



[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.56.

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